Laplace's equation in 1D

For a potential that depends only on x, $\nabla^2\phi = 0 \quad \Rightarrow \quad \frac{d^2\phi}{dx^2} = 0 \quad \Rightarrow \quad \phi(x) = mx + c$. If $c = 0$, then $\phi(x) = mx$.

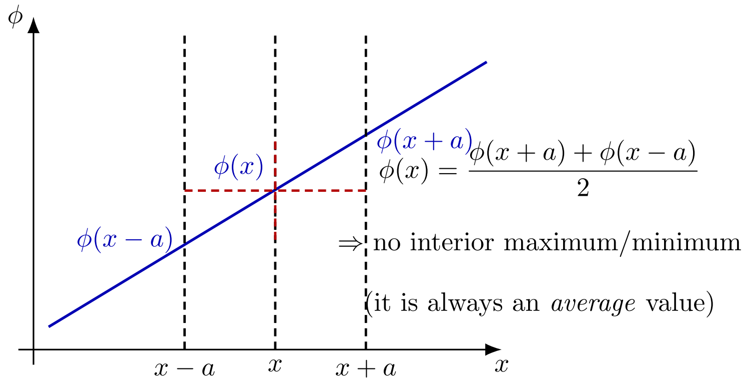


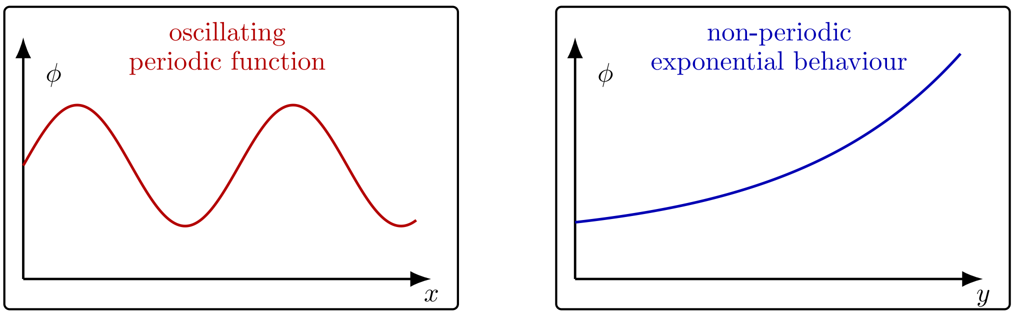
Figure 1: In 1D, Laplace $\Rightarrow$ linear $\phi(x)$, hence $\phi(x)$ equals the average of neighboring values.

Laplace's equation in 2D

$$\nabla^2\phi = 0 \quad \Rightarrow \quad \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0. \tag{1}$$

So the curvatures must *balance*:

$$\frac{\partial^2\phi}{\partial x^2} < 0 \Rightarrow \frac{\partial^2\phi}{\partial y^2} > 0, \qquad \text{or} \qquad \frac{\partial^2\phi}{\partial x^2} > 0 \Rightarrow \frac{\partial^2\phi}{\partial y^2} < 0.$$



Laplace forces opposite signs of curvature:
$\phi_{xx} = -\phi_{yy}$

Figure 2: In 2D, solutions can show periodic behaviour in one direction and exponential in the other (via separation), while still satisfying $\phi_{xx} + \phi_{yy} = 0$.

Saddle point picture (local shape under Laplace)

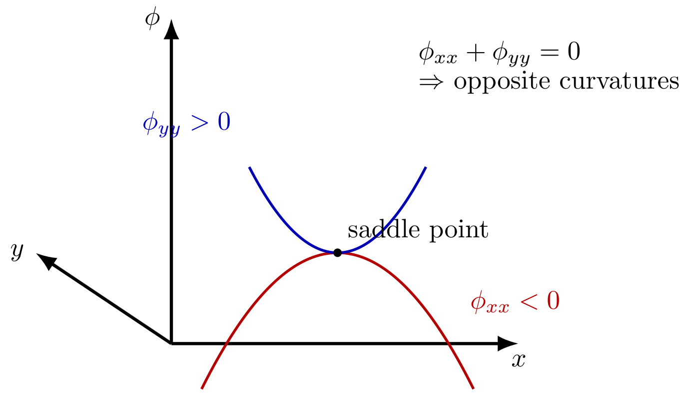


Figure 3: Local geometry of a 2D harmonic function: concave down in one direction and concave up in the perpendicular direction (saddle).

Key note: saddle point and "no interior maxima/minima"

Saddle point (local picture). If $\nabla^2\phi = 0$ at an interior point, then $\phi_{xx} = -\phi_{yy}$. Hence the surface bends *up* in one direction and *down* in another $\Rightarrow$ the point is locally a **saddle**.

Mean-value / maximum principle connection (1D & 2D).

- In 1D, $\phi'' = 0 \Rightarrow \phi$ is linear, so $\phi(x) = \big(\phi(x+a) + \phi(x-a)\big)/2$ and there is *no interior* maximum/minimum (unless ϕ is constant).
- In 2D (and higher), harmonic functions satisfy a mean-value property (average over a small circle/sphere equals the center value). Therefore, a strict interior maximum/minimum is impossible; **extrema occur on the boundary** (unless ϕ is constant everywhere).

General solution for Eqn. (1) (separation of variables)

Assume a *separable* form

$$\phi(x, y) = X(x)\,Y(y). \tag{2}$$

Substitute into Laplace's equation in 2D,

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0,$$

to get

$$\frac{\partial^2}{\partial x^2}\big(X(x)Y(y)\big) + \frac{\partial^2}{\partial y^2}\big(X(x)Y(y)\big) = 0, \tag{3}$$

$$Y(y)\,\frac{d^2X(x)}{dx^2} + X(x)\,\frac{d^2Y(y)}{dy^2} = 0. \tag{4}$$

Divide (4) by $X(x)Y(y)$ (valid where $X, Y \neq 0$):

$$\frac{1}{X(x)}\frac{d^2X(x)}{dx^2} + \frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2} = 0. \tag{5}$$

Define

$$F(x) \triangleq \frac{1}{X(x)}\frac{d^2X(x)}{dx^2}, \qquad G(y) \triangleq \frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2}.$$

Then (5) becomes

$$F(x) + G(y) = 0 \qquad \text{(must hold for every } (x, y)\text{)}. \tag{6}$$

Why must $F(x)$ and $G(y)$ be constants? Fix any $y = y_0$. Then from (6),

$$F(x) = -G(y_0) \quad \text{for all } x.$$

The right-hand side is a *constant* (since y_0 is fixed), hence $F(x)$ must be constant for all x. Similarly, fixing any $x = x_0$ shows $G(y)$ must be constant for all y.

Equivalently (checking sample points): if at $x = 0$, suppose $F(0) = 10$, then (6) forces $G(y) = -10$ for all y. But if at $x = 1$, $F(1) = 11$, then $11 + (-10) \neq 0$, contradicting (6). So $F(x)$ cannot depend on x; it must be a constant, and likewise $G(y)$.

$$\boxed{F(x) = \text{constant}, \qquad G(y) = \text{constant}}$$

and since $F(x) + G(y) = 0$,

$$\boxed{F(x) = C, \qquad G(y) = -C}.$$

We denote that constant (conveniently) by $+k^2$:

$$\boxed{\frac{1}{X}\frac{d^2X}{dx^2} = +k^2, \qquad -\frac{1}{Y}\frac{d^2Y}{dy^2} = +k^2} \tag{7}$$

so that

$$\frac{d^2X}{dx^2} - k^2X = 0, \tag{8}$$

$$\frac{d^2Y}{dy^2} + k^2Y = 0. \tag{9}$$

Solutions. From (8),

$$\boxed{X(x) = C_1 e^{kx} + C_2 e^{-kx}} \tag{10}$$

(exponential / non-periodic along x). From (9),

$$\boxed{Y(y) = C_3 \cos(ky) + C_4 \sin(ky)} \tag{11}$$

(oscillatory / periodic along y).
Therefore the separated solution is

$$\boxed{\phi(x,y) = X(x)Y(y) = \left(C_1 e^{kx} + C_2 e^{-kx}\right)\left(C_3 \cos(ky) + C_4 \sin(ky)\right).} \tag{12}$$

Why do we write the constant as $-k^2$ or $+k^2$?

- The separation constant can be any real constant C. Writing it as $\pm k^2$ is just a **convenient re-parameterization** (so that k has units of inverse-length and remains real).
- If we choose $C = +k^2$, we get

$$X'' - k^2 X = 0 \ \text{(exponentials)}, \qquad Y'' + k^2 Y = 0 \ \text{(sin/cos)}.$$

- If instead we choose $C = -k^2$, the roles swap:

$$X'' + k^2 X = 0 \ \text{(sin/cos)}, \qquad Y'' - k^2 Y = 0 \ \text{(exponentials)}.$$

- The **sign** is picked to match the boundary conditions (periodic vs decaying/growing behaviour in a given direction). The case $k = 0$ is the special linear case ($X'' = 0$ and $Y'' = 0$).

(Annotation / remark.) In Cartesian coordinates, separation typically produces *sines/cosines* in one direction and *exponentials* in the other (as above). In other coordinate systems, the separated ODEs change form: Example

$$\text{cylindrical } (\rho, \phi): \ \text{radial equation } \Rightarrow \text{Bessel functions},$$

Laplace's 2D problem: Rectangular pipe (example)

Two infinitely-long (along z) grounded metal plates are placed at $y = 0$ and $y = a$ (so $\phi = 0$ on these walls). The side walls at $x = \pm b$ are formed by metal strips maintained at a constant potential V_0. A thin insulating layer at each corner prevents electrical contact (shorting) between the grounded plates and the V_0 strips. Find the potential $\phi(x,y)$ inside the rectangular region $-b < x < b,\ 0 < y < a$.

Governing equation (inside):

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \qquad \text{for } -b < x < b,\ 0 < y < a.$$

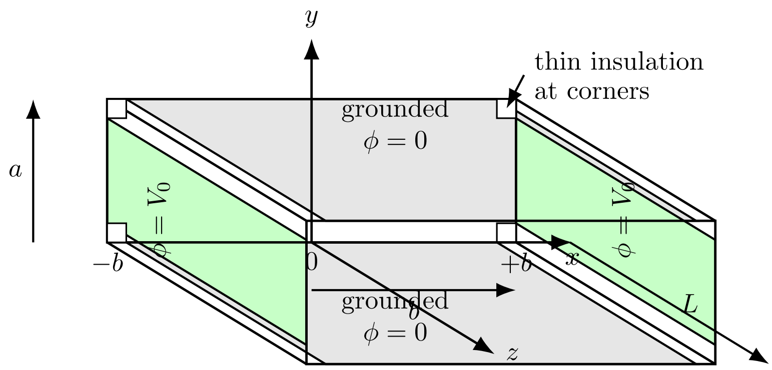


Figure 4: Rectangular pipe (uniform along z): grounded plates at $y = 0$ and $y = a$; side walls at $x = \pm b$ held at V_0; thin insulation at the corners prevents shorting.

Solution (set-up and first simplification)

Step 0: 2D Laplace equation. Since the configuration is uniform along the z-direction (infinitely long pipe),

$$\frac{\partial}{\partial z}(\cdot) = 0 \quad \Longrightarrow \quad \nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0, \qquad -b < x < b,\ 0 < y < a.$$

Boundary conditions (B.C.):

$$\phi(x,0) = 0, \qquad \phi(x,a) = 0, \tag{13}$$

$$\phi(-b,y) = V_0, \qquad \phi(b,y) = V_0. \tag{14}$$

Why is this a 2D Laplace problem? (intuition)

The pipe is infinitely long (or uniform) along z. So nothing changes as we move along $z \Rightarrow$ no z-variation in ϕ. Hence the Laplacian reduces to the 2D form: $\phi_{xx} + \phi_{yy} = 0$.

Step 1: Separation of variables. Try $\phi(x,y) = X(x)Y(y)$. Substitution gives

$$\frac{X''}{X} = -\frac{Y''}{Y} = k^2,$$

so that

$$X'' - k^2 X = 0, \qquad Y'' + k^2 Y = 0.$$

Hence the separated solutions are

$$X(x) = C_1 e^{kx} + C_2 e^{-kx}, \qquad Y(y) = C_3 \sin(ky) + C_4 \cos(ky), \tag{15}$$

and

$$\phi(x,y) = \big(C_1 e^{kx} + C_2 e^{-kx}\big)\big(C_3 \sin(ky) + C_4 \cos(ky)\big). \tag{16}$$

Step 2: Use symmetry about $x = 0$. Since the left and right walls are at the *same* potential, $\phi(b, y) = \phi(-b, y) = V_0$, the geometry and boundary conditions are symmetric with respect to the plane $x = 0$. Therefore,

$$\boxed{\phi(x, y) = \phi(-x, y)} \qquad \text{(even symmetry in } x\text{).} \tag{17}$$

Apply (17) to (16):

$$(C_1 e^{kx} + C_2 e^{-kx})\, Y(y) = (C_1 e^{-kx} + C_2 e^{kx})\, Y(y).$$

For nontrivial $Y(y)$,

$$C_1 e^{kx} + C_2 e^{-kx} = C_1 e^{-kx} + C_2 e^{kx} \quad \Rightarrow \quad (C_1 - C_2)\big(e^{kx} - e^{-kx}\big) = 0.$$

Since $e^{kx} - e^{-kx} \neq 0$ for general x, we must have

$$\boxed{C_1 = C_2} \tag{18}$$

and thus

$$X(x) = C_1\big(e^{kx} + e^{-kx}\big) = 2C_1 \cosh(kx).$$

Absorbing the factor $2C_1$ into a new constant A,

$$\boxed{X(x) = A \cosh(kx)} \tag{19}$$

so the separated form becomes

$$\phi(x, y) = \cosh(kx)\Big(C_3' \sin(ky) + C_4' \cos(ky)\Big). \tag{20}$$

Why does symmetry force $\cosh(kx)$? (intuition)

Equal potentials on $x = +b$ and $x = -b$ mean the solution must be **even** in x. In the general x-dependence $C_1 e^{kx} + C_2 e^{-kx}$, the **even** combination is $e^{kx} + e^{-kx} = 2\cosh(kx)$, while the **odd** combination is $e^{kx} - e^{-kx} = 2\sinh(kx)$. Even symmetry $\Rightarrow$ keep $\cosh(kx)$ and eliminate $\sinh(kx)$.

Apply boundary conditions at $y = 0$ and $y = a$

BC (i): grounded plates at $y = 0$ and $y = a$.

$$\phi(x, 0) = 0, \qquad \phi(x, a) = 0.$$

At $y = 0$:

$$\phi(x, 0) = \cosh(kx)\Big(C_3' \sin 0 + C_4' \cos 0\Big) = \cosh(kx)\, C_4'.$$

Since $\cosh(kx) > 0$ for all x, the only way to satisfy $\phi(x, 0) = 0$ for every x is

$$\boxed{C_4' = 0}. \tag{21}$$

Hence

$$\phi(x,y) = C_3' \cosh(kx)\sin(ky). \tag{22}$$

Why must $C_4' = 0$? (intuition)

At $y = 0$, the bracket becomes just C_4'. Also $\cosh(kx)$ never becomes zero (it is always positive). So the only way $\phi(x,0)$ can be zero for *all* x is to set the coefficient of $\cos(ky)$ to zero.

At $y = a$:

$$\phi(x,a) = C_3' \cosh(kx)\sin(ka) = 0 \qquad \forall x.$$

Again $\cosh(kx) > 0$. For a non-trivial solution ($C_3' \neq 0$), we must have

$$\sin(ka) = 0 \quad \Longrightarrow \quad \boxed{k = \frac{n\pi}{a}, \quad n = 1, 2, 3, \dots}. \tag{23}$$

Why does k become discrete? (intuition- but will be able to appreciate when you study electromagnetic waves)

The potential must be zero on *both* grounded boundaries $y = 0$ and $y = a$. This forces the y-dependence to be a standing wave that has nodes at both ends:

$$\sin(ky) = 0 \text{ at } y = 0 \text{ and } y = a \;\Rightarrow\; ka = n\pi.$$

So only specific (discrete) spatial frequencies fit inside the interval $0 < y < a$.

Therefore each allowed mode is

$$\phi_n(x,y) = C_n \cosh\left(\frac{n\pi x}{a}\right) \sin\left(\frac{n\pi y}{a}\right), \qquad n = 1, 2, \dots \tag{24}$$

Superposition (linearity of Laplace's equation)

If ϕ_1 and ϕ_2 are solutions of Laplace's equation in the region, then any linear combination

$$\alpha\phi_1 + \beta\phi_2 \qquad (\alpha, \beta \in \mathbb{R})$$

is also a solution (because ∇^2 is a linear operator).

Why can we add solutions? (intuition)

Laplace's equation is linear:

$$\nabla^2(\alpha\phi_1 + \beta\phi_2) = \alpha\nabla^2\phi_1 + \beta\nabla^2\phi_2.$$

If both $\nabla^2\phi_1 = 0$ and $\nabla^2\phi_2 = 0$, then the combination also gives 0.

Hence the most general solution satisfying the grounded conditions at $y = 0$ and $y = a$ is the series

$$\boxed{\phi(x,y) = \sum_{n=1}^{\infty} C_n \cosh\!\left(\frac{n\pi x}{a}\right) \sin\!\left(\frac{n\pi y}{a}\right)} \tag{25}$$

Apply boundary condition at $x = \pm b$ and determine coefficients

From the previous section,

$$\phi(x,y) = \sum_{n=1}^{\infty} C_n \cosh\!\left(\frac{n\pi x}{a}\right) \sin\!\left(\frac{n\pi y}{a}\right), \qquad 0 < y < a,\ -b < x < b. \tag{26}$$

BC (ii): $\phi(\pm b, y) = V_0$ for $0 < y < a$. Using (26) at $x = b$,

$$V_0 = \sum_{n=1}^{\infty} C_n \cosh\!\left(\frac{n\pi b}{a}\right) \sin\!\left(\frac{n\pi y}{a}\right). \tag{27}$$

Define

$$A_n \triangleq C_n \cosh\!\left(\frac{n\pi b}{a}\right), \tag{28}$$

so the boundary condition becomes a sine series on $(0, a)$:

$$\boxed{V_0 = \sum_{n=1}^{\infty} A_n \sin\!\left(\frac{n\pi y}{a}\right)} \tag{29}$$

Why a sine Fourier series here? (intuition)

Because the grounded boundaries enforce $\phi(x,0) = \phi(x,a) = 0$, the natural eigenfunctions in y are $\sin\!\left(\frac{n\pi y}{a}\right)$ (they vanish at $y = 0, a$). Therefore *any* reasonable function prescribed on $0 < y < a$ (including the constant V_0) can be expanded in this sine basis.

Use orthogonality to find A_n. Multiply (29) by $\sin\!\left(\frac{m\pi y}{a}\right)$ and integrate from 0 to a:

$$\int_0^a V_0 \sin\!\left(\frac{m\pi y}{a}\right) dy = \sum_{n=1}^{\infty} A_n \int_0^a \sin\!\left(\frac{n\pi y}{a}\right) \sin\!\left(\frac{m\pi y}{a}\right) dy.$$

Using

$$\int_0^a \sin\!\left(\frac{n\pi y}{a}\right) \sin\!\left(\frac{m\pi y}{a}\right) dy = \begin{cases} 0, & m \neq n, \\ \frac{a}{2}, & m = n, \end{cases} \tag{30}$$

we obtain

$$A_m\left(\frac{a}{2}\right) = \int_0^a V_0 \sin\left(\frac{m\pi y}{a}\right) dy. \tag{31}$$

Evaluate the integral:

$$\begin{aligned}\int_0^a V_0 \sin\left(\frac{m\pi y}{a}\right) dy &= V_0 \left[-\frac{a}{m\pi} \cos\left(\frac{m\pi y}{a}\right)\right]_0^a \\ &= \frac{V_0 a}{m\pi}\Big(1 - \cos(m\pi)\Big).\end{aligned} \tag{32}$$

Substitute into (31):

$$A_m = \frac{2V_0}{m\pi}\Big(1 - \cos(m\pi)\Big). \tag{33}$$

Since $\cos(m\pi) = (-1)^m$,

$$1 - \cos(m\pi) = \begin{cases} 0, & m \text{ even}, \\ 2, & m \text{ odd}. \end{cases}$$

Hence

$$\boxed{A_n = \begin{cases} 0, & n \text{ even}, \\ \dfrac{4V_0}{n\pi}, & n \text{ odd}. \end{cases}} \tag{34}$$

Recover C_n from A_n. Using (28),

$$C_n = \frac{A_n}{\cosh\left(\frac{n\pi b}{a}\right)}. \tag{35}$$

Final potential inside the pipe

Substitute (34) and (35) into (26). Only odd n remain:

$$\boxed{\phi(x, y) = \frac{4V_0}{\pi} \sum_{n=1,3,5,\ldots}^{\infty} \frac{1}{n} \frac{\cosh\left(\frac{n\pi x}{a}\right)}{\cosh\left(\frac{n\pi b}{a}\right)} \sin\left(\frac{n\pi y}{a}\right)} \tag{36}$$

Take-away (summary)

- Uniform along $z \Rightarrow$ solve 2D Laplace: $\phi_{xx} + \phi_{yy} = 0$ in $-b < x < b,\ 0 < y < a$.
- Grounded plates at $y = 0, a$ enforce $\sin\left(\frac{n\pi y}{a}\right)$ eigenfunctions and quantize $k = \frac{n\pi}{a}$.
- Equal side-wall potentials at $x = \pm b$ force even symmetry in $x \Rightarrow \cosh\left(\frac{n\pi x}{a}\right)$.
- Matching $\phi(\pm b, y) = V_0$ reduces to a sine Fourier series of a constant; only odd n survive.
- Final answer:

$$\phi(x, y) = \frac{4V_0}{\pi} \sum_{n=1,3,5,\ldots}^{\infty} \frac{1}{n} \frac{\cosh\left(\frac{n\pi x}{a}\right)}{\cosh\left(\frac{n\pi b}{a}\right)} \sin\left(\frac{n\pi y}{a}\right).$$